STATISTICAL STRUCTURED PREDICTION

Question set (Part 1)

December 2025

Administrative issues

- This quiz represents the 30% of the final score of PEE.
- Each question has a score according to its difficulty.
- Answers can be handwritten, as long as this is readable.
- Please present the results in a readable "PDF" document and upload the file to the open task in poliformat.
- Deadline for delivering these questions is **22 December 2025**

Question 1 (2 point)

Demonstrate with a concrete example that a weighted regular grammar that is not proper can generate a probabilistic language. Consider a grammar that can generate infinite strings (hint: see here). Your grammar should be able to analyse strings from Σ^* where $\Sigma = \{a, b\}$.

Apply the method described in Section IV of the previous cite to transform your weighted regular grammar in an equivalent proper grammar.

Question 2 (2 points)

Compute the 5-best solutions with both models of the previous question to a string with length 6. The selected string should have at least 5 different paths.

Question 3 (2 points)

Suppose an instructor wants to determine whether a student has understood the class material based on the exam score. For this purpose, he has the following Bayesian network defined by the factorization associated with its joint probability.

$$P(I, H, U, E) = P(I)\ P(H)\ P(U \mid I, H)\ P(E \mid U)\,,$$

Where:

I	represents whether the student is very intelligent (yes or no)
H	represents whether the student is very hardworking (yes or no)
U	represents whether the student has understood the class material (yes or no)
E	represents whether the student has obtained a high score on the exam (yes or no)

The probabilities are,

$P(I)$

yes	0.3
no	0.7

$P(H)$

yes	0.4
no	0.6

$P(U \mid I, H)$

I	H	yes	no
yes	yes	0.9	0.1
yes	no	0.6	0.4
no	yes	0.5	0.5
no	no	0.1	0.9

$P(E \mid U)$

U	yes	no
yes	0.9	0.1
no	0.2	0.8

What is the probability that a student who obtained a high exam score did not understand the class material?

Question 4 (1.0 points)

As introduced in class, a CRF can be viewed as a factor graph where the bigram transition scores can be expressed as:

$$\Psi_t\,(y_{t-1},\ y_t,\ x_t) = \exp\,\{\sum_{k=1}^{K}\ \theta_k\ f_k\,(y_{t-1},\ y_t,\ x_t)\,\}$$

We also saw that the Forward algorithm for CRFs could be used to calculate the score of an observation, $x_1 \dots x_t$ ending at $y_t = s \in \mathcal{Y}$ as,

$$\alpha_t(s) \quad \stackrel{\text{def}}{=} \quad \sum_{y_1^t;\, y_t=s} \ \prod_{i=1}^{t}\ \Psi_i\,(y_{i-1},\ y_i,\ x_i)$$

In this question, we will consider a *"trigram"* model for CRF, where the transition, $\Psi_t(\,\cdot\,)$, takes the form,

$$\Psi_t\,(y_{t-2}, y_{t-1},\ y_t,\ x_t) \ = \ \exp\,\{\sum_{k=1}^{K}\ \theta_k\ f_k\,(y_{t-2}, y_{t-1},\ y_t,\ x_t\,\}\,.$$

$\Psi_t(\,\cdot\,)$ is again a local transition score. We have assumed in this definition that $y_{-1} = y_0 = \#$. We call it a trigram CRF because we now consider sequences of three output labels (y_{t-2}, y_{t-1}, y_t).

Taking as reference the Forward algorithm seen in class, give a new version of this Forward algorithm for a *trigram* CRF.

Question 5 (3 points)

Given the CKY *Inside* algorithm writte a similar algorithm to compute the number of different parse trees that a string can have assuming the the SCFG is given. Provide an example for a simple grammar with at least 3 non-terminals and with a string with size at least 5 terminals. The number of parse trees has to be larger than 2.